Exercise 1. Prove that the polynomial equation

$$\frac{x^{1121}}{1121} + \frac{x^{2021}}{2021} + x + 1 = 0$$

has exactly one real root.

Proof. Let $f : \mathbb{R} \to \mathbb{R}$. Assume, for the purpose of proof, that

$$\forall x \in \mathbb{R} : f(x) = \frac{x^{1121}}{1121} + \frac{x^{2021}}{2021} + x + 1.$$

We want to prove that f has at least one root in $\mathbb{R}$. Given f is polynomial, f is continuous. Compute, for the purpose of proof,

$$f(-2020) = 1 - 2020 - \frac{2020^{1121}}{1121} - \frac{2020^{2021}}{2021} < 0$$

and

$$f(2020) = 1 + 2020 + \frac{2020^{1121}}{1121} + \frac{2020^{2021}}{2021} > 0.$$

By the Intermediate Value Theorem, $\exists c \in (-2020, 2020)$ such that

$$f(c) = \frac{c^{1121}}{1121} + \frac{c^{2021}}{2021} + c + 1 = 0.$$

Thus, f has at least one root in $\mathbb{R}$, as needed for proof. We want to prove that f has at most one root in $\mathbb{R}$. Given f is polynomial, f is differentiable. By the Differentiation Rules,

$$f'(x) = \frac{d}{dx}\left(\frac{x^{1121}}{1121} + \frac{x^{2021}}{2021} + x + 1\right) = x^{1120} + x^{2020} + 1.$$

Compute, for the purpose of proof,

$$\forall x \in \mathbb{R} : f'(x) = x^{1120} + x^{2020} + 1 > 0 + 0 + 1 = 1 > 0.$$

By the Derivative Characterization of Monotone Functions, f is strictly increasing on $\mathbb{R}$. Thus, f has at most one root in $\mathbb{R}$, as needed for proof. □

Exercise 2. Compute the fourth Taylor polynomial for:

(a) $f(x) = \sin x$ at $x = 0$.

(b) $g(x) = (1 - x)^{-1}$ at $x = -1$.

Proof. Let $f(x) := \sin x$ and $g(x) := (1 - x)^{-1}$.

(a) By the Derivative Computation of Trigonometric Functions,

$$f'(x) = \cos x \wedge f''(x) = -\sin x \wedge f'''(x) = -\cos x \wedge f^{(4)}(x) = \sin x.$$

Thus,

$$f_4(x) = 0 + 1(x - 0) + 0(x - 0)^2 - \frac{1}{6}(x - 0)^3 + 0(x - 0)^4 = x - \frac{x^3}{6},$$

as needed.

(b) By the Chain Rule of Differentiable Functions,

$$g'(x) = \frac{1}{(1-x)^2} \wedge g''(x) = \frac{2}{(1-x)^3} \wedge g'''(x) = \frac{6}{(1-x)^4} \wedge g^{(4)}(x) = \frac{24}{(1-x)^5}.$$

Thus,

$$g_4(x) = \frac{1}{2} + \frac{1}{4}(x+1) + \frac{1}{8}(x+1)^2 + \frac{1}{16}(x+1)^3 + \frac{1}{32}(x+1)^4,$$

as needed.

□

Exercise 3. Compute:

(a)

$$\lim_{x\to 0} \frac{x - \sin x}{x^3}.$$

(b)

$$\lim_{x\to \frac{\pi}{2}} \frac{1 - \sin x}{(x - \frac{\pi}{2})^2}.$$

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$.

(a) Choose $f(x) = x - \sin x$ and $g(x) = x^3$. Then f and g are differentiable and continuous. By the Differentiation Rules of Functions,

$$f'(x) = \frac{d}{dx}(x - \sin x) = 1 + \cos x \wedge g'(x) = \frac{d}{dx}(x^3) = 3x^2.$$

By the Limit Characterization of Continuity,

$$\lim_{x\to 0} f(x) = f(0) = 0 - \sin 0 = 0 \wedge \lim_{x\to 0} g(x) = g(0) = 0^3 = 0.$$

By the L'Hôpital Rule for Indeterminate Forms,

$$\lim_{x\to 0} \frac{x - \sin x}{x^3} = \lim_{x\to 0} \frac{f(x)}{g(x)} = \lim_{x\to 0} \frac{f'(x)}{g'(x)} = \lim_{x\to 0} \frac{1 + \cos x}{3x^2} = \infty.$$

(b) Choose $f(x) = 1 - \sin x$ and $g(x) = (x - \pi/2)^2$. Then f and g are twice differentiable and continuous. By the Differentiation Rules of Functions,

$$f'(x) = \frac{d}{dx}(1 - \sin x) = -\cos x \wedge g'(x) = \frac{d}{dx}\left(x - \frac{\pi}{2}\right)^2 = 2x - \pi$$

and

$$f''(x) = -\frac{d}{dx}(\cos x) = \sin x \wedge g''(x) = \frac{d}{dx}(2x - \pi) = 2.$$

By the Limit Characterization of Continuity,

$$\lim_{x\to \frac{\pi}{2}} f(x) = f\left(\frac{\pi}{2}\right) = 1 - \sin\frac{\pi}{2} = 0 \wedge \lim_{x\to \frac{\pi}{2}} g(x) = g\left(\frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^2 = 0$$

and

$$\lim_{x\to\frac{\pi}{2}} f'(x) = f'\left(\frac{\pi}{2}\right) = -\cos\frac{\pi}{2} = 0 \wedge \lim_{x\to\frac{\pi}{2}} g'(x) = g'\left(\frac{\pi}{2}\right) = \pi - \pi = 0.$$

By the L'Hôpital Rule for Indeterminate Forms,

$$\lim_{x\to\frac{\pi}{2}} \frac{1-\sin x}{(x-\frac{\pi}{2})^2} = \lim_{x\to\frac{\pi}{2}} \frac{f(x)}{g(x)} = \lim_{x\to\frac{\pi}{2}} \frac{f'(x)}{g'(x)} = \lim_{x\to\frac{\pi}{2}} \frac{f''(x)}{g''(x)} = \lim_{x\to\frac{\pi}{2}} \frac{\sin x}{2} = 1.$$

□

Exercise 4. Suppose that $f : (a, b) \to \mathbb{R}$ is three times continuously differentiable, $c \in (a, b)$, $f'(c) = f''(c) = 0$, and $f'''(c) > 0$. Prove that f has a neither a local maximum nor a local minimum at c.

Proof. Let $f : (a, b) \to \mathbb{R}$ have three continuous derivatives. Suppose, for the purpose of proof, $c \in (a, b)$, $f'(c) = f''(c) = 0$ and $f'''(c) > 0$. By Taylor's Theorem, $\forall x \in [a, b]$ $\exists y \in (x, c) \cup (c, x)$ such that

$$\begin{aligned} f(x) &= f(c) + f'(c)(x-c) + \frac{f''(c)}{2}(x-c)^2 + \frac{f'''(y)}{6}(x-c)^3 \\ &= f(c) + 0(x-c) + 0(x-c)^2 + \frac{f'''(y)}{6}(x-c)^3 = f(c) + \frac{f'''(y)}{6}(x-c)^3. \end{aligned}$$

By the Pointwise Characterization of Continuity,

$$\lim_{x\to c} f'''(x) = f'''(c) > 0 \implies \exists\delta > 0\ \forall y \in (c-\delta, c+\delta) : f'''(y) > 0.$$

If $x \in (c, c+\delta)$, then $\exists y \in (c, x)$ such that

$$f(x) = f(c) + \frac{f'''(y)}{6}(x-c)^3 > f(c) + 0 = f(c).$$

If $x \in (c-\delta, c)$, then $\exists y \in (x, c)$ such that

$$f(x) = f(c) + \frac{f'''(y)}{6}(x-c)^3 < f(c) + 0 = f(c)$$

Thus, f has neither a local minimum nor a local maximum at c, as needed of proof. □

Exercise 5. Let $a < b$, and define a sequence of tagged partitions by

$$\underline{x}^{(r)} := \left\{ a + (b-a)\frac{k}{r} : k = 0, ..., r \right\}$$

and

$$\underline{\xi}^{(r)} := \left\{ a + (b-a)\frac{k}{r} : k = 1, ..., r \right\}.$$

(a) Compute $\|\underline{x}^{(r)}\|$.

(b) Let $f(x) = \alpha x + \beta$. Prove that

$$\lim_{r\to\infty} S_f\left(\underline{x}^{(r)}, \underline{\xi}^{(r)}\right) = \alpha\frac{b^2-a^2}{2} + \beta(b-a).$$

You may not use the Fundamental Theorem of Calculus.

Proof. Let $f : [a,b] \to \mathbb{R}$. Define a sequence of tagged partitions by

$$\underline{x}^{(r)} := \left\{a + (b-a)\frac{k}{r} : k = 0, ..., r\right\}$$

and

$$\underline{\xi}^{(r)} := \left\{a + (b-a)\frac{k}{r} : k = 1, ..., r\right\}.$$

(a) Suppose, for the remainder of proof, $a < b$. Compute, for the purpose of proof, that

$$\begin{aligned}\|\underline{x}^{(r)}\| &= \max\left\{\left(a + (b-a)\frac{k}{r}\right) - \left(a + (b-a)\frac{k-1}{r}\right) : k = 1, ..., r\right\} \\ &= \max\left\{(b-a)\frac{1}{r} : k = 1, ..., r\right\} = \frac{b-a}{r}.\end{aligned}$$

(b) Suppose, for the remainder of proof, $f(x) = \alpha x + \beta$. Compute, for the purpose of proof, that

$$\begin{aligned}S_f\left(\underline{x}^{(r)}, \underline{\xi}^{(r)}\right) &= \sum_{k=1}^{r}\left[\alpha\left(a + (b-a)\frac{k}{r}\right) + \beta\right]\frac{(b-a)}{r} \\ &= \sum_{k=1}^{r}\left[(\alpha a + \beta)\frac{(b-a)}{r} + \alpha k\frac{(b-a)^2}{r^2}\right] \\ &= (\alpha a + \beta)(b-a)\sum_{k=1}^{r}\frac{1}{r} + \alpha(b-a)^2\sum_{k=1}^{r}\frac{k}{r^2} \\ &= (\alpha a + \beta)(b-a) + \frac{\alpha}{2}(b-a)^2\left(1 + \frac{1}{r}\right) \\ &= \alpha(ab - a^2) + \frac{\alpha}{2}(b^2 - 2ab + a^2) + \beta(b-a) + \frac{\alpha}{2r}(b-a)^2 \\ &= \alpha\frac{b^2-a^2}{2} + \beta(b-a) + \alpha\frac{(b-a)^2}{2r}.\end{aligned}$$

By the Algebraic Properties of Convergent Functions,

$$\begin{aligned}\lim_{r\to\infty} S_f\left(\underline{x}^{(r)}, \underline{\xi}^{(r)}\right) &= \lim_{r\to\infty}\left[\alpha\frac{b^2-a^2}{2} + \beta(b-a) + \alpha\frac{(b-a)^2}{2r}\right] \\ &= \alpha\frac{b^2-a^2}{2} + \beta(b-a) + \alpha\frac{(b-a)^2}{2}\lim_{r\to\infty}\frac{1}{r} \\ &= \alpha\frac{b^2-a^2}{2} + \beta(b-a) + 0 = \alpha\frac{b^2-a^2}{2} + \beta(b-a),\end{aligned}$$

as needed for proof.

□